



Chemometrics Web App: Second Order Calibration

Comprehensive Step-by-Step User Guide & Tutorial Manual

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1. Introduction & overview

The Chemometrics Web App: Second Order Calibration (CWA: SOC) is an open-source, browser-accessible, and desktop-supported graphical platform designed to perform multi-way tensor decomposition, second-order calibration, and analyte quantitation in analytical chemistry datasets.

In modern analytical laboratories, hyphenated and multi-dimensional instruments generate second-order data matrices for each sample. Examples include Excitation-Emission Fluorescence Matrices (EEM), Liquid Chromatography paired with Diode Array Detection (LC-DAD), and Kinetic-Spectroscopic measurements. When analyzing real-world complex samples, such as environmental water, biological fluids, food matrices, or industrial formulations, target analytes are frequently accompanied by unknown, uncalibrated chemical interferents.

Traditional first-order calibration techniques require that all potential interferents be present in the calibration set. In contrast, CWA: SOC exploits the mathematical property known as the Second-Order Advantage. By modeling multi-way data using advanced tensor algorithms like Parallel Factor Analysis (PARAFAC) and Multivariate Curve Resolution - Alternating Least Squares (MCR-ALS), CWA: SOC resolves clean spectral signatures and quantifies target compounds even when unmodeled interferents overlap with analyte signals.

CWA: SOC is fully integrated into the Chemometrics Web App (CWA) platform developed by LTAP-UERJ in international collaboration with UNL, UNR, and CONICET. It follows CWA: Data Handling & Preprocessing (<https://doi.org/10.1016/j.chemolab.2022.104696>), CWA: Exploratory Analysis (<https://doi.org/10.1016/j.chemolab.2023.104810>) and CWA: Synthetic Sampling.

2. Software access & download links

CWA: SOC is accessible online via browser cloud and as a standalone desktop executable package for Windows:

Access Mode	Link / Destination
Cloud Web Application	https://ltap.shinyapps.io/second_order_calibration/
Standalone Desktop (.exe, Windows)	https://drive.google.com/drive/folders/1l8dB4BGKVjqPMrvA5ZCLvafRNtTOCFew?usp=drive_link
Official LTAP Laboratory Portal	https://www.ltapuerj.com.br/
INPI Software Registration Mirror	https://sites.google.com/view/ltap-uerj/cwa
GitHub Source Code & Assets	https://github.com/LTAP-UERJ

Note on Performance: For large, high-dimensional tensor arrays (e.g., EEM excitation-emission matrices with hundreds of samples or high-density chromatographic landscapes), the standalone desktop executable is recommended to bypass free-tier cloud memory limits and session timeout limits imposed by shinyapps.io.

3. System requirements & architecture

Cloud Version: Compatible with any modern web browser (Google Chrome, Mozilla Firefox, Microsoft Edge, Safari) on Windows, macOS, Linux, Android, or iOS. No software installation is required.

Standalone Desktop Version: Windows 10/11 (64-bit). Minimum 4 GB RAM (8 GB recommended for multi-sample 3D tensor arrays). Includes an isolated, pre-configured R runtime environment.

Computational Architecture: CWA: SOC relies on a modular, high-performance R backend:

Component	Technology / Package	Role in CWA: SOC
User Interface	R/Shiny + shinydashboard	Graphical rendering, sidebar menu, reactive event handlers.
Data Visualization	plotly, ggplot2, DT	Interactive 2D contours, 3D surface landscapes, spectral profile plots.
Matrix & Tensor Math	pracma, abind, ptw	Numerical integration, 3D array stacking, Parametric Time Warping.
Multi-Way Modeling	multiway, eemR, staRdom	PARAFAC trilinear decomposition, CORCONDIA, EEM scatter removal.
Model Diagnostics	ggResidpanel, GGally, ellipse	Residual analysis, leverage vs orthogonal distance plots, confidence ellipses.

3.1 Operational modes comparison: Standalone desktop executable vs. Cloud web application

Chemometricians traditionally operate within local desktop mathematical environments (such as MATLAB®, RStudio®, or compiled standalone executable toolboxes), performing analytical routines directly on local workstation hardware and handling data files via native operating system directories. To bridge traditional desktop workflows with modern web application convenience, CWA: SOC provides two distinct operational deployment modes: the Cloud Web Application (hosted on shinyapps.io) and the Standalone Desktop Executable (.exe for Windows).

Both operational modes execute identical underlying chemometric processing engines, including PARAFAC multi-way decomposition, MCR-ALS optimization, Rayleigh and Raman scattering removal, interactive 2D/3D Plotly visualizations, model diagnostics (CORCONDIA, residual maps), and pseudo-univariate concentration quantitation. However, understanding their computational differences, memory management models, and data security profiles allows analysts to choose the optimal execution environment for their operational requirements:

1. Cloud Web Application (ShinyApps.io)

- **Concept & Terminology:** The app runs in cloud server containers managed over remote cloud infrastructure. "Cloud computing" indicates that computational routines and memory allocation occur on high-performance remote servers accessed seamlessly through standard web browsers (Google Chrome, Mozilla Firefox, Microsoft Edge, Safari).
- **Key Advantages:** Does not require any software installation or local library dependency management; universal cross-platform compatibility across Windows, macOS, Linux, ChromeOS, iOS, and Android; instant, seamless access to the latest algorithmic updates without manual maintenance; highly recommended for teaching, academic workshops, rapid exploratory testing, and workstations with restricted administrative software installation rights.

- **Limitations & Operational Constraints:** Standard free-tier cloud containers operate under fixed server RAM caps (~500 MB to 4 GB RAM allocation) and enforce automated idle session timeouts (~15–30 minutes of user inactivity automatically reinitializes server memory). Processing large multi-sample 3D tensor arrays (for example, datasets with 50+ Excitation-Emission Fluorescence Matrices or high-density LC-DAD chromatographic-spectral landscapes) may reach container memory limits or experience network transfer delays during dataset upload.

2. Standalone Desktop Executable (.exe)

- **Concept & Terminology:** A self-contained Windows desktop package (.exe) that operates 100% locally on the host computer. It features a bundled, isolated R runtime environment, eliminating any need for pre-installed R/RStudio software or external package installations.
- **Key Advantages:** Unrestricted local hardware utilization, directly leverages all available system RAM (for example, 8 GB, 16 GB, 32 GB+) and multi-core CPU processing power without cloud server RAM caps or session timeout resets. High-dimensional 3D multi-way datasets process significantly faster with zero session interruption risk. Absolute data privacy and confidentiality, sensitive industrial, commercial, food, or pharmaceutical analytical matrices remain entirely local on the host computer. Operates fully offline in secure analytical laboratory PCs or field laptops without internet connectivity.
- **Limitations & Operational Constraints:** Native package restricted to 64-bit Windows operating systems (Windows 10/11); requires downloading the installer package (~400 MB) and use approximately 1 GB of local disk space.

Comparison between ShinyApps and executable standalone interface:

Operational Feature / Dimension	Cloud Web Application (ShinyApps.io)	Standalone Desktop Executable (.exe)
Software Access & Setup	Immediate web browser access; zero installation or setup	Downloadable portable package with pre-bundled R runtime
Supported Operating Systems	Universal (Windows, macOS, Linux, ChromeOS, iOS, Android)	Windows 10 / 11 (64-bit)
Computation & Hardware	Remote cloud server container	Local host CPU & system RAM
RAM & Tensor Capacity	Server-capped container memory (~500 MB – 4 GB)	Unrestricted local system RAM (8 GB, 16 GB, 32 GB+)
Session Stability & Timeouts	Automated idle timeout (~15–30 min inactivity resets session)	Persistent, unlimited local session (No timeouts)
Internet Requirement	Active internet connection required	100% offline operational capability
Data Security & Confidentiality	Data processed in cloud RAM over encrypted HTTPS	100% local processing (Data never leaves host PC)
Primary Recommended Use Cases	Quick exploratory runs, teaching, academic workshops, cross-platform users	Large multi-sample 3D tensors, offline lab PCs, confidential data

4. Software navigation & core modules

The CWA: SOC interface is organized sequentially into six main workflow tabs. Analyst is indicating to proceed from left to right:

Recommended Workflow: Tab 1: Upload Files → Tab 2: Preprocessing → Tab 3: Compare & Accept → Tab 4: PARAFAC → Tab 5: MCR-ALS → Tab 6: Information, Acknowledgments & References

4.1 Tab 1: Data Import & Axes Configuration

Upload calibration matrices, test matrices, and corresponding blank files (.csv, .xlsx, .xls, .txt). Configurable axis mapping allows analysts to define B-mode (Emission) and C-mode (Excitation) orientation.

Requirement	Specification
Calibration Files	Upload 2D matrices representing calibration samples.
Test Files	Upload 2D matrices representing target test samples containing interferents.
Blank Files	Upload single or multiple blank matrices for automated background subtraction.
Mode Orientation	Select whether wavelengths are arranged along columns or rows (B-mode vs C-mode).

4.2 Tab 2: Preprocessing & Scattering Removal

Raw EEM landscapes contain physical scattering artifacts (Raman and Rayleigh scattering) that break trilinearity and corrupt PARAFAC decomposition. Tab 2 provides a complete preprocessing toolkit:

- **Blank Correction:** Automated subtraction of single or averaged blank matrices from sample matrices.
- **Wavelength Slicing:** Trimming Excitation (B-mode) or Emission (C-mode) wavelength boundaries to exclude non-informative or noisy regions.
- **Scattering Removal:** Simultaneous removal of 1st Order Rayleigh, 2nd Order Rayleigh, 1st Order Raman, and 2nd Order Raman scattering lines with custom bandwidths.
- **Missing Value Interpolation:** Spline, linear 2D, or zero-imputation over excised scattering channels to restore continuous matrix surfaces.
- **Baseline Correction:** Asymmetric Least Squares (ALS)
- **Peak Alignment:** Parametric Time Warping (PTW) and iCoshift for baseline drift compensation. Beware when use this peak alignment preprocessment because it was originally made for first order data and may input artifacts on second order datasets.

4.3 Tab 3: Compare & Accept Diagnostic

Before model construction, Tab 3 displays side-by-side 2D contour maps and 3D surface projections comparing raw EEM matrices against preprocessed landscapes, allowing visual confirmation of scatter removal.

4.4 Tab 4: PARAFAC Modeling & Pseudo-Univariate Calibration

PARAFAC decomposes the 3D data tensor X (Samples $\times$ B-mode $\times$ C-mode) into three matrices A , B , and C :

$$\text{Trilinear Model: } X_{ijk} = \sum (a_i * b_j * c_k) + e_{ijk}$$

- **Constraints:** Enforce Non-negativity, Smoothness, Orthogonality, Unimodality, or Monotonicity per mode to ensure physically meaningful factor profiles.
- **Model Rank Verification:** Evaluate CORCONDIA (Core Consistency Diagnostic). A value $> 85\%$ indicates a valid trilinear structure; values dropping below 60% indicate over-factoring.
- **Pseudo-Univariate Calibration:** Fit linear regression between resolved sample scores (A-mode) and reference concentrations to predict target analytes in test samples.

4.5 Tab 5: MCR-ALS Modeling & Quantitation

MCR-ALS decomposes data matrices under bilinear assumptions, making it ideal for systems with non-trilinear shifts (for example, chromatographic retention time shifts):

- **Initial Guess:** Derive pure variable initial guesses using Pure (algorithm based on SIMPLISMA).
- **ALS Resolution:** Iterative optimization under Non-negativity, Unimodality, and Closure constraints until convergence (std error $< 0.1\%$).
- **Second-Order Quantitation:** Quantify target analytes in test matrices containing uncalibrated interferents.

For both PARAFAC and MCR-ALS it is possible to export resolved score matrices, B-mode/C-mode profiles, predicted analyte concentrations, statistical metrics (RMSEP, REP%, R^2), and Analytical figure of merits such as sensitivity, specificity, analytical sensitivity, limit of detection, limit of quantification.

5. Algorithm parameter reference guide

5.1 Preprocessing Parameters

Parameter	Type	Default	Description
raman1N / raman2N	Integer	15	Bandwidth (nm) for 1st and 2nd Order Raman scatter excision.
rayleigh1N / rayleigh2N	Integer	15	Bandwidth (nm) for 1st and 2nd Order Rayleigh scatter excision.
typeN	Select	Spline (1)	Interpolation method: 0 = Set NA to 0; 1 = Spline; 2 = Wavelength-wise; 4 = Linear 2D.
exmin / exmax	Numeric	Auto	Wavelength limits for Excitation trimming.
emmin / emmax	Numeric	Auto	Wavelength limits for Emission trimming.

5.2 PARAFAC Parameters

Parameter	Type	Default	Description
nfactors	Integer	3	Number of PARAFAC components (factors) to extract.
const	Select	Non-negative	Mode constraint combination (e.g., non-negativity, unimodality, smoothness).
norm	Select	Fmax	Scaling norm for factor profiles (Fmax, Frobenius, 1-norm, Spectral 2-norm).
maxit	Integer	2500	Maximum iterations for Alternating Least Squares convergence.
ctol	Float	1e-6	Convergence tolerance for relative error change.

6. Diagnostic interpretation guide: acceptance and rejection criteria

Diagnostic Tool	Acceptable Criteria (Pass)	Problematic Criteria (Reject)
CORCONDIA (PARAFAC)	Value $\geq 85\text{--}90\%$ indicates a robust, valid trilinear decomposition.	Value $< 60\%$ indicates model rank oversaturation or invalid trilinearity.
Residual Map	Flat, unstructured noise residuals across all EEM channels.	Structured residual peaks matching analyte excitation/emission lines.
Spectral Loadings	Smooth, physically interpretable non-negative emission/excitation profiles.	Oscillatory, negative, or severe peak splitting artifacts.
Regression Line (R^2)	$R^2 \geq 0.98$ for pseudo-univariate concentration calibration.	$R^2 < 0.90$ indicating non-linear response or incorrect component matching.
Prediction (REP%)	Error REP% approximately 5.0% for test sample concentration predictions.	REP% $> 10.0\%$ indicating unresolved interference or poor factor stability.

7. Step-by-Step guided tutorials with demo files

Tutorial 1: EEM Fluorescence Dataset (PARAFAC Calibration)

Objective: The example contains 2 analytes and a single interferent in samples test1.txt to test5.txt, and 2 analytes and 2 interferents in samples test6.txt to test10.txt. Sensor data: 321 360 1 261 290 1 (emission-excitation).

- **Step 1 — Data Import:** Navigate to Tab 1 'Upload Files'. Load the demo calibration and test EEM matrices by click on upload and import the excel files.
- **Step 2 — Scatter Removal:** Navigate to Tab 2 'Preprocessing'. Enable 'Remove Scattering' (Raman 1=5nm, Rayleigh 1=5nm), navigate to Interpolation of missing values, enable Missing values are interpolated within data and select 'Spline Interpolation'. Do the same for test samples.
- **Step 3 — Inspection:** Navigate to Tab 3 'Compare & Accept'. Verify that diagonal Rayleigh scatter lines have been completely removed and interpolated.
- **Step 4 — Decomposition:** Navigate to Tab 4 'PARAFAC'. Set Components = 3, Constraint = 'Non-negative' for all A, B and C modes. Click 'Run PARAFAC'.

- **Step 5 — Diagnostic Verification:** Check CORCONDIA score. Inspect resolved B-mode and C-mode spectral profiles at models plots and noise at Metrics.
- **Step 6 — Quantitation:** Open 'Pseudo-Univariate Calibration' tab. Select target analyte concentrations for both calibration and test samples each one on their respective tab. Review regression line and AFOMS. Export predicted concentrations.

Tutorial 2: LC-DAD Dataset (MCR-ALS Second-Order Advantage)

Objective: The example contains 2 analytes and a single interferent in samples test.

- **Step 1 — Data Upload:** Upload 2D chromatographic-spectral matrices exactly like for PARAFAC example.
- **Step 2 — Initial Guess:** Navigate to Tab 5 'MCR-ALS'. Click 'Unrestricted Initial Solution', click on Augment data, go to PCA tab to estimate the number of components and finally to pure profile estimation to estimate the initial guess bases on pure variable analysis (Pureness threshold = 10%).
- **Step 3 — ALS Optimization:** Go to MCR tab, select Non-negativity constraint for concentration (C) and spectral (S) profiles and unimodality to Augmented mode. Keep trilinearity with No once there is retention time shifts and select the interferents profile. Click 'Run MCR-ALS'.
- **Step 4 — Profile Matching:** Verify convergence. Match resolved spectral profile against reference pure spectrum.
- **Step 5 — Quantitation:** Go to Calibration tab, load the calibration and test samples concentration for target analyte in test samples. Confirm Second-Order Advantage recovery. Review regression line and AFOMS. Export predicted concentrations.

8. Chemometric best practices for Second Order Calibration

- **1. Preprocessing Priority:** Direct preprocessing on second-order data should generally be minimized; however, for excitation-emission fluorescence matrices (EEM), always remove 1st and 2nd order Rayleigh and Raman scattering lines prior to PARAFAC modeling to prevent artificial component generation. Apply spline or 2D linear interpolation to fill excised scatter channels. Exercise caution when applying peak alignment tools (such as PTW or iCoshift), as algorithms designed for first-order data can introduce structural artifacts into multi-way tensor datasets.
- **2. Second-Order Advantage Scope:** The Second-Order Advantage allows accurate quantification of target analytes in complex samples containing unmodeled chemical interferents. However, the target analyte itself MUST be present in the calibration set to establish valid pseudo-univariate regression.
- **3. Model Rank Determination & Trilinearity Verification:** Always combine multiple diagnostics to determine the correct number of components:
 - **Core Consistency (CORCONDIA):** Aim for values $\geq 85\text{-}90\%$ to confirm a valid trilinear structure. Values below 60% signal over-factoring or structural violations.

- **Residual Analysis:** Inspect residual maps to ensure flat, unstructured random noise rather than residual peaks matching excitation/emission profiles.
- **Spectral Profiles:** Confirm that resolved B-mode and C-mode loadings are smooth and physically interpretable.
- **4. Physical Constraint Selection:** Apply appropriate feasible constraints during modeling to eliminate rotational ambiguity and ensure realistic profiles:
 - Enforce **Non-negativity** across B-mode and C-mode spectral channels for fluorescence emission/excitation and absorption spectra.
 - Use **Unimodality** along chromatographic retention dimensions in MCR-ALS when single chromatographic peaks are expected.
- **5. PARAFAC vs. MCR-ALS Algorithm Selection:**
 - Use **PARAFAC** when datasets adhere strictly to trilinear proportions (for example, EEM fluorescence).
 - Switch to **MCR-ALS** when soft bilinear modeling is required, such as in liquid chromatography (LC-DAD) datasets exhibiting run-to-run retention time shifts that violate strict trilinearity.
 - **6. Comprehensive Model Validation & Figures of Merit:** Evaluate calibration performance beyond basic R^2 values (≥ 0.98). Check the Relative Error of Prediction ($REP \leq 5.0\%$) and calculate Analytical Figures of Merit (AFOMs), including sensitivity, analytical sensitivity, limit of detection (LOD), limit of quantification (LOQ), and specificity, to confirm model reliability.

8.1 Workflow-Oriented Methodological Guide & Results Interpretation

To assist both novice and experienced chemometricians, CWA: SOC structures second-order multivariate calibration into an intuitive, sequential workflow. Below is an exhaustive stage-by-stage guide detailing the underlying mathematical operations, calculated parameters, and their analytical interpretation:

- Stage 1 — Data Import & Flexible Formatting:
 - Input Options: Users may import multi-way matrices with standard Mode 1 (rows) and Mode 2 (columns) labels, or utilize the newly introduced 'Import without labels' feature. In label-free mode, the application automatically synthesizes equidistant coordinate vectors based on user-defined first value, last value, and channel count. Matrix formats supported include .xlsx, .xls, .csv, and .txt.
 - Concentration Vectors: Calibration and test response files can now be loaded in .xlsx, .xls, .csv, or .txt format, with or without a 'Concentration' column header.
 - Missing Data (NaN) Detection: Automatic scanning alerts analysts to any NaN/missing cells upon upload, preventing downstream ALS divergence.
- Stage 2 — Pre-processing & Baseline Correction:
 - Rayleigh & Raman Scattering: Excises 1st and 2nd order scattering ridges in EEM fluorescence. Excised channels are reconstructed using cubic spline or 2D linear interpolation to preserve trilinearity.

- Eilers Baseline Correction: Implements penalized asymmetric least squares baseline adjustment to eliminate background spectral slopes and detector baseline drifts.

- Stage 3 — Model Building (PARAFAC & MCR-ALS):

- PARAFAC: Decomposes the 3-way array under strict trilinearity constraints. Settings are divided into 'Basic Modeling Parameters' (number of components, non-negativity constraints) and 'Advanced Settings' (convergence criteria, initialization mode).

- MCR-ALS: Resolves bilinear components in the presence of run-to-run retention time shifts or non-trilinear profiles. Initial estimates are derived via pure-variable SIMPLISMA or SVD.

- Stage 4 — Model Assessment & Diagnostics:

- Core Consistency (CORCONDIA): Evaluates the appropriateness of the trilinear model. Values $> 85\%$ confirm trilinear validity; values $< 60\%$ indicate over-factoring or structural breakdown.

- Residual Diagnostics: Divided into explicit 'Normality Assessment' (Q-Q plot and count-aligned histogram) and 'Homoscedasticity Assessment' (residuals vs. fitted values and integer-indexed observation plot) to verify classical OLS assumptions.

- Stage 5 — Pseudo-Univariate Calibration & Validation:

- Formatted Regression Output: Raw R console dumps are replaced with structured Bootstrap/HTML tables displaying Slope, Intercept, Standard Errors, t-values, p-values, R^2 , and Residual Standard Error.

- Recovery Accuracy t-Test: A formal one-sample Student's t-test tests $H_0: \mu = 100\%$ nominal recovery, evaluating whether predictions exhibit statistically significant systematic bias.

- Elliptical Joint Confidence Region (EJCR): Calculates the joint 95% confidence region for slope and intercept. The theoretical point (1, 0) must fall within the ellipse to confirm analytical accuracy.

- Stage 6 — Analytical Figures of Merit (AFOMs):

- Sensitivity (SEN): Slope of the pseudo-univariate analyte response function.

- Analytical Sensitivity (γ): Ratio of sensitivity to instrumental noise ($\gamma = \text{SEN} / s_x$), defining the inverse of the minimum concentration difference distinguishable at 95% confidence.

- Limit of Detection (LOD) & Limit of Quantification (LOQ): Calculated per IUPAC/ISO recommendations taking into account calibration uncertainty, leverage (h_0), and second-order selectivity.

- Selectivity (SEL) & %REP: Degree of analyte signal orthogonality to interferents, and Relative Error of Prediction across external test samples.

8.2 Pedagogical Datasets: Benchmark vs. Realistic Experimental Variability

To maximize pedagogical efficacy, CWA: SOC provides two complementary classes of example datasets:

1. Benchmark Datasets (01_EEFM_Rayleigh_Olivieri & 02_LC_DAD_Olivieri): Synthetic/benchmark mixtures with exceptionally high signal-to-noise ratios. Recoveries typically approach $\sim 100.0\%$, producing very small EJCR ellipses that demonstrate theoretical perfection but may exclude (1, 0) due to tiny numerical offsets.

2. Realistic Experimental Datasets (03_EEFM_Realistic_Variability & 04_LC_DAD_Realistic_Variability): Feature realistic laboratory noise (~1.5–1.8% proportional noise, baseline drift). These datasets produce realistic recovery spreads (96–104%), suitably sized EJCRC ellipses covering (1, 0), and informative residual diagnostic plots, ideal for teaching and methodological validation.

9. Troubleshooting & Frequently Asked Questions (FAQ)

Q: Why does PARAFAC return negative spectral loadings?

A: Fluorescence emission/excitation and UV-Vis absorption profiles cannot have negative intensities. In Tab 4 (PARAFAC Modeling), ensure that the constraint selector for B-mode and C-mode is explicitly set to 'Non-negative'.

Q: My CORCONDIA score is 0% (or below 60%). What happened?

A: A low CORCONDIA score indicates a severe violation of the trilinear assumption required by PARAFAC. Common causes include:

- Un-removed Rayleigh or Raman scattering lines.
- Retention time shifts across sample runs in chromatographic datasets.
- Extracting too many components.

Severe matrix-induced spectral shifts (consider switching to MCR-ALS in Tab 5).

Q: The cloud web app shows a memory limit error, close or resets during processing.

A: Free-tier cloud server containers enforce strict memory limits (~500 MB to 4 GB RAM) and automated idle session timeouts (15–30 minutes). When processing high-density 3D tensor arrays (for example, 50+ EEM matrices or high-resolution LC-DAD landscapes), download and run the standalone desktop executable package (.exe) for 64-bit Windows from Google Drive to utilize your workstation's full local RAM capacity.

Q: MCR-ALS fails to converge or produces unrealistic multi-peak spectra.

A: This typically stems from poor initial profile estimates or under-constrained optimization. In Tab 5 (MCR-ALS):

- Derive initial pure-variable guesses using SIMPLISMA (Pure) with an appropriate pureness threshold (for example, 10%).
- Apply Non-negativity constraints to concentration (C) and spectral (S) modes, and apply Unimodality along augmented modes where applicable.

Q: I am getting high prediction errors (REP > 10%) or poor calibration linearity ($R^2 < 0.90$).

A: Verify that the target analyte peak/factor has been correctly identified and matched in the resolved profiles. Ensure that background blank matrices were properly subtracted in Tab 2 and that unmodeled interferents are not co-extracted within the target factor.

Q: Tab 1 gives an error when importing calibration or test matrices.

A: Verify that your file format (.csv, .xlsx, .xls, .txt) aligns with the requested matrix orientation. Ensure that B-mode (Emission) and C-mode (Excitation) axes map consistently across all calibration, test, and blank files, and confirm that all matrices have identical dimensions (the same number of channels/variables) in both B-mode and C-mode.

10. Citation, Software Information & Acknowledgments

Software Version & Distribution

- **Software Name: Chemometrics Web App: Second Order Calibration (CWA: SOC)**
- **Version: 1.0 (Release Date: August 2026)**
- **Cloud Access Portal:** https://ltap.shinyapps.io/second_order_calibration/
- **Standalone Windows Executable on GitHub or drive:** <https://github.com/LTAP-UERJ?tab=repositories>
https://drive.google.com/drive/folders/1l8dB4BGKVjqPMrvA5ZCLvafRNtTOCFew?usp=drive_link
- **Official Laboratory Website:** <https://www.ltapuerj.com.br/>
- **INPI Registration Mirror:** www.ltapuerj.com.br/cwa

Primary Citation: If CWA: SOC is used in research, please cite the primary software registration and paper:

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